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SUPERSEDING
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FEDERAL SPECIFICATION

CAMERAS, TELEVISION, MONOCHROME, CLOSED CIRCUIT (GENERAL SPECIFICATION)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the general requirements for monochrome closed circuit television cameras. Detailed requirements for a particular type of camera are specified in a series of detailed specifications (see 3.1 and 6.3).

1.2 Classification.

1.2.1 Types. The types of television cameras covered by this specification are as follows:

Type I - General Purpose Camera.
Type II - High Resolution Camera, selectable scan rates.
Type III - Studio Camera with electronic viewfinder.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

J-C-175 - Cable Assembly, Power, Electrical.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-170 - Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252 - Wired Equipment, Classification of Visual and Mechanical Defects For.

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(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc. (UL) Standards:

UL 198 - Fuses.
UL 512 - Fuseholders.
UL 796 - Printed Circuit Boards.

(Copies may be obtained from the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Ill. 60611; 1285 Walt Whitman Road, Melville, Long Island, New York 11746; or 1655 Scott Boulevard, Santa Clara, Calif. 95050.)

Electronic Industries Association (EIA) Standards:

RS-170 - Monochrome Television Studio Facilities.
RS-330 - Electrical Performance Standards for Closed Circuit Television Cameras 525/60 Interlaced 2:1.
RS-343 - Electrical Performance Standards for High Resolution Monochrome Closed Circuit Television Cameras.
EIA Resolution Chart - 1956 with Gray Scale Overlay Strips.
EIA Linearity Chart.
EIA Closed Circuit Television Definitions, Industrial Electronics Bulletin No. 1.

JEDEC Publications:

No. 52 - Electron Tube Registration List
No. 53 - Semiconductor Device Registration List

(Copies may be obtained from the Electronic Industries Association, Engineering Department, 2001 I Street, N. W., Washington, D. C. 20006.)

American National Standards Institute, Inc. (ANSI) Standards:

C99.1 - 1966 - Criteria for Inspection for Highly Reliable Soldered Connections in Electronic and Electrical Applications.

International Electrotechnical Commission Recommendations:

IEC 68-1 (1968) - Basic Environmental Testing Procedures for Electronic Components and Electronic Equipment Part 1: General.
IEC 68-2-1 (1966) - Part 2: Tests - Test A: Cold.
IEC 68-2-2 (1966) - Part 2: Tests - Test B: Dry Heat.
IEC 68-2-6 (1966) - Part 2: Tests - Test F: Vibration.
IEC 68-2-13 (1966) - Part 2: Tests - Test M: Low Air Pressure.

(Copies may be obtained from the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

Institute of Electrical and Electronics Engineers (IEEE) Standards:

No. 202 Television: Aspect Ratio and Geometric Distortion, Methods of Measurement of
No. 205 Television: Luminance Signal Levels, Measurement of
No. 208 Video Techniques: Resolution of Camera Systems, Measurement of

(Copies or reproductions may be obtained from the Institute of Electrical and Electronics Engineers, 345 East 47 Street, New York, N. Y. 10017).

3. REQUIREMENTS

3.1 Detailed specification. The individual item requirements shall be as specified herein and in accordance with the applicable detailed specification (see 1.1 and 6.3). In the event of any conflict between this specification and the applicable detailed specification, the latter shall govern.

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3.2 General. Each individual camera shall be complete when delivered and shall have been inspected and tested for conformance to the requirements as specified herein and in the applicable detailed specification (see 3.1 and 6.3). The camera shall have all the devices and controls necessary for full automatic operation requiring only to be plugged into a local source of ac power to operate. The camera shall be new and of the manufacturer's standard commercial design or model current at the time of delivery, except for such minor modification from the manufacturer's standard commercial design or model as may be required to conform to the requirements as specified herein and in the applicable detailed specification.

3.3 Fire, casualty, and electrical hazards. Each contractor shall submit to the contracting agency proof that the fuses, fuseholders, and printed circuit boards, of the television camera he proposes to supply under this specification, conform to the requirements of the Underwriters' Laboratories, Inc., UL 198, Fuses; UL 512, Fuseholders; and UL 796 Printed Circuit Boards. The UL label, or listing mark will be accepted as evidence that the fuses, fuseholders, and printed circuit boards of the television camera conform to this requirement.

3.3.1 In lieu of the UL label, or listing mark, the contractor shall submit independent proof, satisfactory to the contracting agency, that the fuses, fuseholders, and printed circuit boards conform to the applicable requirements of the published standards including methods of tests of the Underwriters' Laboratories Standards UL 198, UL 512, and UL 796.

3.3.2 Compliance with the above preliminary requirements in regard to fire, casualty, and electrical hazard does not absolve the contractor from complete compliance with the other requirements of this specification in order to secure acceptance of his products.

3.4 Preproduction model. Unless otherwise specified (see 6.1) a preproduction model shall be submitted for examination, test, and inspection in accordance with section 4, to determine compliance with the requirements as specified herein and in the applicable detailed specification. In all respects not specifically covered in the specification, the preproduction model shall represent the identical quality of materials, workmanship, and design which will be used in the fabrication of the production units. The approved preproduction model shall be used as a guide to inspect production units and may be presented for acceptance as the last unit of the contract.

3.5 Design and construction. Each camera shall incorporate printed circuit modular construction throughout. The design of the cameras shall provide for easy replacement of the printed circuit modules. The electronic circuits shall utilize solid state components with the following permissable exceptions, the camera tube; the video preamplifier, which may be a sub-miniature tube; and the high voltage rectifier and display tube of any viewfinder that may be specified as part of the camera. Printed circuit boards shall have plated through holes and shall comply with the requirements of UL 796. Each camera shall be constructed to provide rigid support for its electrical and optical systems so that unintentional changes in alignment or microphonic effects will not occur during camera movement or lens adjustment. At the same time, the camera shall be as lightweight and compact as possible. The camera tube and any other devices subject to burnout or short operating life, shall be mounted to facilitate replacement. The camera shall have effective shielding on the yoke assembly to protect the focus, deflection and alignment coils, and the camera tube from external magnetic fields. To maintain a standard of quality and reliability, all components shall be conservatively rated, circuit design shall incorporate safety margins of not less than 25 percent, where possible, with respect to dissipation ratings, voltage ratings, and current carrying capacity.

3.5.1 Stability. The circuits used in the cameras shall have been designed to insure continuous operation in accordance with the requirements specified herein regardless of component aging.

3.5.2 Test points. Conveniently accessible test points shall be provided where knowledge of the characteristics of a significant waveform or electrical value is necessary in the adjustment, measurement, and observation of operating parameters that verify performance requirements. All test points shall be identified with a TP number.

3.5.2.1 Video calibration. Each camera shall be provided with a video calibration test jack. The test jack shall allow insertion of a test signal in place of the camera tube output, to calibrate the gain of the video amplifier stages, and permit proper setting of the camera tube current.

3.5.3 Input and output connectors. Each camera shall accept male 75 ohm BNC or UHF type signal connectors. If a camera, as manufactured, is equipped to accept other type signal connectors and is provided with "between series" adapters to accommodate male 75 ohm BNC or UHF connectors, it shall be acceptable.

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3.6 Parts and material. The parts, material, and processes used in the construction of a particular type camera, but not definitely specified, shall be equivalent to and interchangeable with the corresponding part, material, or process in the manufacturer's normal commercial product of that type. "Normal commercial product" shall be interpreted to mean an end item(s) covered by this specification. The Government reserves the right to require any bidder to submit, for approval of the contracting officer, detailed engineering drawings to proposed end items or components differing from the "normal commercial products or standard component(s) "prior to and as a condition precedent to the award of the contract.

3.7 Nonselection of parts. Specified requirements for each type camera shall be obtained with standard available commercial electronic components, EIA registered tubes and transistors, and without selection of parts. Nonselection means that having determined that the parts meet their detailed specification or other detailed requirements to which they are manufactured no further consideration is given to their characteristics when they are assembled into the camera. Nonselection precludes matching of parts or selective assemblies. Nonselection also precludes the necessity of interchange of parts during or after mounting in the camera.

3.8 Safety and protective devices. Adequate safeguards shall be provided to protect personnel from exposure to high voltage during operation or adjustment. Each camera shall operate safely and in accordance with requirements herein and in the applicable detailed specification with either side of the power source line grounded.

3.9 Power supply. Each camera shall operate in accordance with the requirements specified herein and in the applicable detail specification, from a normal power source of 105 to 130 volts 60 Hz ac, single phase. Each camera shall operate on regulated dc power provided by internal power supplies.

3.9.1 Power cord. A six foot power cord utilizing three wires, with one of the wires a common ground, shall be supplied as part of each camera. The cord shall be in accordance with Type I Series S-J of J-C-175, except that the color of the cord may be black, brown, or gray.

3.9.2 Fuses, fuseholders. Each camera shall be protected from circuit overloads by a fuse or fuses in the power source line. The fuseholder shall be of a type to facilitate replacement of the fuse. The fuses and fuseholders shall meet the requirement of UL 198 and UL 512.

3.10 Aspect ratio. Unless otherwise specified in the detailed specification, the aspect ratio shall be 4 to 3.

3.11 Geometric distortion. When measured in accordance with 4.5.2, no picture element shall be displaced from its true position, referred to the subject, by more than the value specified in the applicable detailed specification.

3.12 Resolution.

3.12.1 Horizontal limiting resolution. The horizontal limiting resolution for a particular type camera, when measured in accordance with 4.5.3.1, shall equal or exceed the value specified in the applicable detailed specification.

3.12.2 Horizontal resolution response. The horizontal resolution response for a particular type camera, when measured in accordance with 4.5.3.1, shall equal or exceed the values specified in the applicable detailed specification.

3.12.3 Vertical limiting resolution. The vertical limiting resolution for a particular type camera, when measured in accordance with 4.5.3.2, shall equal or exceed the value specified in the applicable detailed specification.

3.13 Video output. The signal at the video output terminal(s) shall be a composite video signal. The video output signal level for a particular type camera, when measured in accordance with 4.5.4, shall conform to the requirement specified in the applicable detailed specification.

3.13.1 Polarity. Unless otherwise specified in the detailed specification, the polarity of the composite signal at the output terminal of the cameras shall be black negative.

3.13.2 Black level control. Each camera shall have an automatic circuit that will maintain a fixed black level in the output signal, within $\pm$ 5 percent.

3.14 Gray scale. A particular type camera, when measured in accordance with 4.5.5, shall resolve not less than the required shades of gray specified in the applicable detailed specification.

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3.15 Synchronizing signal waveform. The synchronizing waveform of the composite video output signal for a particular type camera, when measured in accordance with 4.5.7, shall conform to the dimensions and tolerances specified in the applicable detailed specification.

3.16 Video amplifier.

3.16.1 Amplitude frequency response. The amplitude frequency response of the video amplifier for a particular type camera, when measured in accordance with 4.5.8, shall conform to the requirement specified in the applicable detailed specification.

3.16.1.1 Aperture correction. Aperture correction for a particular type camera, when measured in accordance with 4.5.9, shall conform to the requirement as specified in the applicable detailed specification.

3.16.2 Low-frequency square-wave response. The low-frequency square-wave response for a particular type camera, when measured in accordance with 4.5.10, shall conform to the requirement specified in the applicable detailed specification.

3.16.3 High-frequency transient response. The high-frequency transient response for a particular type camera, when measured in accordance with 4.5.11, shall conform to the requirement specified in the applicable detailed specification.

3.16.4 Gamma correction. Gamma correction for a particular type camera, when measured in accordance with 4.5.12, shall conform to the requirement as specified in the applicable detailed specification.

3.16.5 Signal-to-noise ratio. When a particular type camera is operated in accordance with the requirements for horizontal limiting resolution as specified in the applicable detailed specification, the ratio of the peak-to-peak white signal (measured with respect to the black level and measured thru a standard "IRE Roll Off" filter) to the RMS noise shall exceed the value specified in the applicable detailed specification.

3.17 Automatic light range. The automatic light range for a particular type camera, when measured in accordance with 4.5.3, shall conform to the requirement as specified in the applicable detailed specification.

3.18 Output impedance. The output impedance shall have a value of 75 ohms ± 5 percent over the video frequency range of the particular type camera and shall be source terminated.

3.18.1 Direct current in the output. The open-circuit direct current and voltage at the output terminal of each camera shall comply with EIA RS-170, paragraph 2.2. The cameras shall be capable of having the output short circuited without damage.

3.19 Hum. The peak-to-peak level of hum appearing across the video output of any camera shall not exceed 1/10 of 1 percent of the blanking-to-reference-white video signal.

3.20 Internally generated transients. There shall be no detectable spurious signals such as "glitches" or "flashes" produced by the operation of any control or switch on the cameras. This shall be true whether a camera is locally or remotely controlled. Periodic transient disturbances, occuring during the picture interval, that are internally generated by the equipment or its power supplies, such as crosstalk, intermodulation, etc., shall not exceed a peak-to-peak value of .005 percent of the nominal peak-to-peak video level.

3.21 Camera tube. The camera tube supplied as part of a particular type camera shall meet the requirements specified herein and in the applicable detailed specification.

3.21.2 Shading. Shading, when measured in accordance with 4.5.6, shall not exceed the value specified in the applicable detailed specification.

3.22 Sweep failure protection. Each camera shall have a camera tube protection circuit, to provide against damage to the camera tube in the event of failure of the horizontal or vertical deflection circuits. The camera tube protection circuit shall be designed to sense both loss of voltage and current.

3.23 Lens mount. Each camera shall have a single lens mount. The lens mounting shall be a 1"-32-thread, "C" mount. The stability of the lens mounting surface in respect to the photosensitive surface of the camera tube shall be such as to preclude differential movement in any plane. The lens mounting surface shall accommodate all standard lenses used for 1-inch vidicon cameras, including motor driven zoom lenses, without requiring additional mechanical support.

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3.23.1 Lens. The lens to be supplied with a particular camera shall be as specified in the procurement document (see 6.1). The type lens to be used shall be determined locally to best suit the scene to be viewed and the camera site. The lens shall be packaged separate from the camera.

3.24 Controls. Controls shall be provided to assure ease of operation and regulation of all specified characteristics. Controls shall be readily accessible without camera disassembly. Where applicable, clockwise rotation of all controls shall result in an increasing function. Primary operating controls shall be positive in action, accurately calibrated, resistant to accidental displacement, and accessible from the rear of the camera and shall conform to the requirements of the applicable detailed specification.

3.25 Operational markings. Controls, indicating devices, input and output connectors and test points shall be clearly labeled to identify the function or intended use. All markings and labels shall be of permanent and legible type.

3.26 Beam focus and alignment. Beam focus shall be electromagnetic with regulating circuitry to assure constant focus current. The focus current shall be regulated to within $\pm$ 1 percent over the range of temperature and line voltage as specified herein. The beam alignment may be electromagnetic or magnetic. Cameras utilizing permanent magnets for beam alignment shall be designed with adequate clamping friction to maintain the magnets stationary under the conditions of vibration and shock specified in 4.5.20.4 and 4.5.20.5.

3.26.1 Focus. Each camera shall be capable of being focused simultaneously over the entire area of the raster.

3.26.2 Optical focus mechanisms. Optical focus mechanisms shall provide smooth positive linkage between focus control and camera tube or lens assembly. Dial cord or other non-positive linkage shall not be used. Non-linear tapered linkage shall provide the same rate of change of focus at all distances at a convenient rate of change of focus. Linkage shall not permit the lens to slip out of focus in any camera position.

3.27 Electronic viewfinder. An electronic viewfinder shall be provided as part of the Type III cameras and shall meet the requirements as specified herein and in the applicable detailed specification.

3.27.1 Resolution. The viewfinder horizontal and vertical limiting resolution, when measured in accordance with 4.5.15.1, shall equal or exceed the values specified in the applicable detailed specification.

3.27.2 Brightness. The viewfinder picture tube brightness, when measured in accordance with 4.5.15.2, shall equal or exceed the value specified in the applicable detailed specification.

3.27.3 Viewfinder geometric distortion. The viewfinder geometric distortion, when measured in accordance with 4.5.15.3, shall not exceed 2 percent of picture height.

3.28 Rough handling. When packaged in accordance with the requirements of section 5, all cameras shall be capable of meeting the applicable inspections of section 4, without subsequent processing, when tested in accordance with 4.5.19.

3.29 Environment. When specified (see 6.1), a particular type camera, when tested in accordance with 4.5.20.1, 4.5.20.2, 4.5.20.3, 4.5.20.4, and 4.5.20.5 for Humidity, Altitude, Temperature, Vibration, and Shock, respectively, shall show no evidence of mechanical defects, corrosion, degradation in performance, below that specified in the applicable detailed specification, for horizontal and vertical limiting resolution, video output signal level, gray scale, and synchronizing signal waveform.

3.30 Instruction manual. The contractor shall supply, with each camera, an instruction manual containing all pertinent electrical and mechanical specifications. The instruction manual shall contain comprehensive operation, installation, and maintenance instructions and shall provide supplemental information such as, schematics, wiring diagrams, and circuit board component location diagrams. Maintenance data shall show circuit waveforms at specific test points throughout the circuit. A complete electrical and mechanical parts list shall be contained in the manual. The electrical parts lists shall contain the components' commercial designation as well as the circuit symbol and electrical ratings. Manufacturers' identification drawings numbers alone will not suffice.

3.31 Workmanship. Each camera, including all parts and accessories, shall be manufactured and finished in such a manner as to meet all the requirements specified herein and in the applicable detailed specification, and shall be free from characteristics or defects which affect the appearance, or which might affect the serviceability or render the camera unsuitable for its intended use. Each camera shall be examined, in accordance with 4.4, for workmanship, and visual and mechanical defects.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Preproduction inspection (4.2.1).
(b) Quality conformance inspection (4.2.2).
(c) Examination of preparation for delivery (see 4.6).

4.2.1 Preproduction inspection. The preproduction inspection shall consist of the appropriate inspections, for the particular type of camera being tested, listed in the group A and B examinations and tests of table I (except for the environmental tests, which will be required only when specified, see 6.1).

4.2.1.1 Sampling for preproduction inspection. Sample size for preproduction inspection shall consist of one complete camera of the particular type the contractor proposes to supply.

4.2.2 Quality conformance inspection. In accordance with 4.1, the contractor shall be responsible for the performance of the quality conformance inspections. A sample number of units, selected in accordance with 4.2.2.1 shall be subjected to the appropriate examinations and tests of table I (see 4.5).

4.2.2.1 Sampling for quality conformance inspection.

4.2.2.1.1 Sampling for group A examinations and tests. Each camera shall be subjected to the appropriate inspections listed in the group A examinations and tests of table I with acceptance based on 4.3.1.

4.2.2.1.2 Sampling for group B examinations and tests. Group B examinations and tests shall be required initially (except that the environmental test shall be required only when specified, see 6.1) and thereafter only when the basic design of the camera or the material of a vital part has been changed. One complete unit, of the type of the contractor proposes to supply, shall be selected and subjected to the group B tests specified in table I, with acceptance based on 4.3.2.

4.3 Acceptance.

4.3.1 Acceptance group A examinations and tests. The results of each examination and test shall be compared with the specified requirements. Failure to conform to the requirements as specified, for any examination or test shall be counted as a defect and the unit shall be rejected. Rejected units, may be offered again for inspection, provided they have been retested for the defect(s) causing rejection and all defects corrected.

4.3.2 Acceptance group B examinations and tests. In the event of failure to conform to the requirements of any group B examination and test, the contractor shall correct the cause of failure in both the test sample and all future production units.

TABLE I. Inspection Tests

Examinations and Tests	Requirement Paragraph	Test Paragraph
Group A		
Visual and mechanical inspection	3.31	4.4
Operating test	3.21, 3.24	4.5.1.2
Aspect ratio	3.10	4.5.2
Horizontal limiting resolution	3.12.1	4.5.3.1
Vertical limiting resolution	3.12.3	4.5.3.2
Video output signal level	3.13	4.5.4
Gray scale	3.14	4.5.5
Shading	3.21.1	4.5.6

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TABLE I. Inspection Tests (Continued)

Examinations and Tests	Requirement Paragraph	Test Paragraph
Group B		
Geometric distortion	3.11	4.5.2
Horizontal resolution response	3.12.2	4.5.3.1
Polarity	3.13.1	4.5.4
Sync signal waveform	3.15	4.5.7
Amplitude frequency response	3.16.1	4.5.8
Aperture correction	3.16.1.1	4.5.9
Low-frequency square-wave response	3.16.2	4.5.10
High-frequency transient response	3.16.3	4.5.11
Gamma correction	3.16.4	4.5.12
Automatic light range	3.17	4.5.13
Internally generated transients	3.20	4.5.14
Viewfinder limiting resolution	3.27.1	4.5.15.1
Viewfinder brightness	3.27.2	4.5.15.2
Viewfinder geometric distortion	3.27.3	4.5.15.3
Voltage range	3.9	4.5.16
Output impedance	3.18	4.5.17
Direct current in output	3.18.1	4.5.18
Rough handling	3.28	4.5.19
(These tests are to be performed only when specified see 6.1)		
Humidity	3.29	4.5.20.1
Altitude	3.29	4.5.20.2
Temperature	3.29	4.5.20.3
Vibration	3.29	4.5.20.4
Shock	3.29	4.5.20.5

4.4 Visual and mechanical inspection. Each camera shall be subjected to a thorough visual inspection for the major defects listed in MIL-STD-252A and the mechanical requirements as specified herein and in the applicable detailed specification. ANSI C99.1-1966 shall be used for inspection criteria when inspecting soldered connections.

4.5 Test methods. Inspections and tests specified herein and in the detailed specifications shall be conducted in accordance with the following methods. Tests shall be performed with the type camera tube that is to be supplied with the particular type camera. Alternate tests may be used provided it can be shown that such tests are in fact adequate to insure compliance with the specification requirements.

4.5.1 Preliminary conditioning. Prior to performing the group A examinations and tests of table I, each camera shall be energized and subjected to an operating period of not less than 48 hours.

4.5.1.1 Test environment. Except for environmental tests (see 3.29), all tests required by this specification shall be made at an ambient temperature of not less than 75°F and at a relative humidity of not less than 45 percent.

4.5.1.2 Operating test. Set up the camera in accordance with 4.5.3 and 4.5.3.1 for horizontal limiting resolution. Adjust the camera for optimum performance. While adjusting the camera, manipulate all electrical and mechanical controls to determine that each control perform its design function, that there is no interaction of controls, and that the controls do not operate at the end of their operating range. Observe the reproduced picture on the monitor. The test pattern arrowheads should just touch the edges of the raster. The picture should be sharp and clear, free of noticeable impairments, such as spots, blemishes, smudge lines, streaks, or a background which is mottled, grainy, or uneven; with no trailing blacks or whites. Tap the camera while observing the picture. The picture should not exhibit abnormal microphonics when the camera is tapped. Vertical or horizontal tilt should not be present.

4.5.2 Aspect ratio and geometric distortion test. Aspect ratio and geometric distortion shall be measured, in accordance with the methods of IEEE No. 202, to determine conformance to the requirements of 3.10 and 3.11.

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4.5.3 Resolution tests. Resolution tests shall be performed with the light level and lens setting specified in the applicable detailed specification.

4.5.3.1 Horizontal limiting resolution and horizontal resolution response. Horizontal limiting resolution and horizontal resolution response shall be measured, in accordance with IEEE No. 208, to determine conformance to the requirements specified in 3.12.1 and 3.12.2. The tests shall be performed using the EIA Resoltuion Chart-1956.

4.5.3.2 Vertical limiting resolution. Vertical limiting resolution shall be measured, in accordance with IEEE No. 208, to determine conformance to the requirement specified in 3.12.3.

4.5.4 Video output signal level test. Set up the camera in accordance with 4.5.3 and 4.5.3.1 for horizontal resolution response. Adjust the camera for optimum performance. Measure the composite video output level in accordance with IEEE No. 205, to determine conformance to the requirements specified in 3.13. Using the oscilloscope display, observe the polarity to determine conformance to the requirement of 3.13.1.

4.5.5 Gray scale test. Set up the camera in accordance with 4.5.3 and 4.5.3.1 for horizontal limiting resolution. Use EIA Resolution Chart-1956 with Gray Scale Overlay Strips, adjust the camera for optimum performance, and evaluate the monitor display subjectively. Observe whether the required shades of gray (as specified in the applicable detailed specification) are easily distinguishable in the reproduced picture (see 3.14).

4.5.6 Shading test. Set up the camera in accordance with 4.5.3 and 4.5.3.1 for horizontal limiting resolution and horizontal resolution response. Adjust the camera for optimum performance. Replace the EIA Resolution Chart with a plain white chart. Use a light meter to check that the chart illumination is uniform. Use the oscilloscope (waveform monitor) to observe the camera video output waveform and adjust the oscilloscope (waveform monitor) for a camera video output presentation of 100 IRE units. Check the video monitor for any obvious shading and use the line select feature of the oscilloscope (waveform monitor) to measure the worst horizontal shading condition noted on the display as the deviation from the 100 IRE units of video signal. Switch the oscilloscope (waveform monitor) to field rate operation and measure the maximum deviation from the 100 IRE units. Determine the amount of shading, measured in percent of peak-to-peak video signal, for conformance to the requirement specified in the applicable detailed specification (see 3.21.1).

4.5.7 Sync signal waveform test. The synchronizing signals shall be measured in accordance with EIA RS-170, EIA RS-330, or EIA RS-343, as applicable, to determine conformance to the requirements specified in the applicable detailed specification (see 3.15).

4.5.8 Amplitude frequency response test. Adjust the camera video amplifier gain to the normal position. Aperture correction, if any, shall be set to zero. Apply a constant level variable frequency sine wave test signal to the camera video preamplifier test input. A video sweep generator may be used for this measurement. Connect a wideband oscilloscope to the video output terminal of the camera. Using the oscilloscope as a measuring device, adjust the frequency and the level of the input test signal to produce an output signal at 100 kHz of 1 volt peak-to-peak. While maintaining the input test signal at this level, vary the test signal over the range of frequencies that is specified in the applicable detailed specification for the particular type camera that is being tested. Measure the response at not less than 10 representative frequency settings. Convert the voltage readings to dB. Compare readings with the requirements specified in the applicable detailed specification (see 3.16.1).

4.5.9 Aperture correction test. Conduct amplitude frequency response test of 4.5.8. Set aperture correction to maximum and repeat the amplitude frequency response test. Compare the results to determine conformance to the requirements specified in the applicable detailed specification (see 3.16.1.1).

4.5.10 Low-frequency square-wave response test. Adjust the video amplifier gain to the normal position. Apply to the camera video preamplifier test input a test signal that consists of a low-frequency square-wave (such as a window signal) synchronized to the field frequency and having a pulse width that measures between 40 to 50 percent of one field time. The test signal must be uniform with no tilt, shading or transient disturbance. Connect a wideband oscilloscope to the video output terminal of the camera. Using the oscilloscope as a measuring device, adjust the level of the input test signal to produce an output video signal of not less than 0.7 volts amplitude measured from reference black to reference white. Sync signals shall not exceed 0.3 volts. Synchronize the oscilloscope to the field frequency and measure any tilt or level shift that may be present to determine conformance to the requirements specified in the applicable detailed specification (see 3.16.2).

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4.5.11 High-frequency transient response test. Adjust the camera video amplifier gain to its normal position. Apply to the camera video preamplifier test input a test signal that consists of a step function transition (such as a rectangular pulse) synchronized to the line scanning frequency and having sufficient duration for steady state to be reached with a rise time that measures less than 50 nanoseconds for type I and type III cameras, and less than 30 nanoseconds for type II cameras. The test signal must be uniform with no tilt, overshoot, ringing or transient disturbance. Connect a wideband oscilloscope to the video output of the camera. Using the oscilloscope as a measuring device, adjust the level of the input test signal to produce an output video signal of not less than 0.7 volts amplitude measured from reference black to reference white. Sync signals shall not exceed 0.3 volts. Synchronize the oscilloscope to the line scanning frequency and measure any ringing or overshoot that occurs just after the transition to determine conformance to the requirements specified in the applicable detailed specification (see 3.16.3).

4.5.12 Gamma correction test. Set up the camera in accordance with 4.5.3 and 4.5.3.1 for horizontal resolution response using the EIA Resolution Chart-1956 with gray scale overlay strips; adjust the camera for optimum performance. Using the line select feature on the oscilloscope (waveform monitor) select a line containing the signal developed by scanning one of the horizontal gray scale overlay strips. Observe the video output of the camera on the oscilloscope (waveform monitor). Adjustment of the gamma control should develop a linear stairstep signal (10 equal amplitude steps) from the log gray scale (see 3.16.4).

4.5.13 Automatic light range test. Set up the camera in accordance with 4.5.3 and 4.5.3.1 for horizontal resolution response; adjust the camera for optimum performance. Observe the video output of the camera using the oscilloscope (waveform monitor) in the line select mode, observing that portion of the test pattern which represents the 10-step gray scale. Adjust the oscilloscope (waveform monitor) for a camera video output presentation of 100 IRE units. With the camera video gain control, set the signal output level to 50 IRE units. Increase the test patterm illumination level until the video level has reached 100 IRE units. Measure the change in illumination level and determine if the automatic light range meets the requirement specified in the applicable detailed specification (see 3.17).

4.5.14 Internally generated transients test. Set up the camera in accordance with 4.5.3 and 4.5.3.1 for horizontal limiting resolution; adjust the camera for optimum performance. Examine all area of the monitor display for the presence of extraneous signals. Cap the camera lens externally to eliminate all light transmission. While manipulating the camera primary operating controls, examine the monitor display for transients such as glitches and flashes to determine conformance to the requirement as specified in 3.20.

4.5.15 Viewfinder. Set up the camera in accordance with 4.5.3 and 4.5.3.1 for horizontal limiting resolution; adjust camera and viewfinder for optimum performance. The viewfinder display shall be sharp and bright with none of the controls at the end of their adjustment range.

4.5.15.1 Viewfinder limiting resolution test. Horizontal and vertical limiting resolution shall be measured in accordance with IEEE No. 208, for conformance to the requirements as specified in the applicable detailed specification (see 3.27.1).

4.5.15.2 Viewfinder brightness test. Follow the procedures of 4.5.15. Advance the brightness control to obtain maximum brightness maintaining good focus. Measure the brightness in foot-lamberts for conformance to the requirement as specified in the applicable detailed specifications (see 3.27.2).

4.5.15.3 Viewfinder geometric distortion test. The geometric distortion shall be measured in accordance with IEEE No. 202, to determine conformance to the requirement specified in 3.27.3.

4.5.16 Voltage range test. Set up the camera in accordance with 4.5.3 and 4.5.3.1; adjust the camera for optimum performance. Connect camera to a variable source of ac that has a range of not less than 105 to 130 volts, 60 Hz. Operate the camera at 105 volts for one hour and at 130 volts for one hour. At the end of each test period, check performance to determine conformance to the requirements for resolution, gray scale, video output, and sync signal tolerance as specified in the applicable detailed specification (see 3.9).

4.5.17 Output impedance test. Output impedance shall be measured in accordance with EIA RS-170 to determine conformance with the requirement as specified in 3.18.

4.5.18 Direct current in output test. Direct current in the output shall be measured in accordance with EIA RS-170 to determine conformance with the requirement as specified in 3.18.1.

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4.5.19 Rough handling test. The camera, packed for normal shipment shall be held not less than 24 inches from a solid concrete floor and dropped as follows:

(a) Once on a bottom corner.
(b) Once on an opposite top corner.
(c) Once on the botton face.

The equipment shall then be unpacked and tested to determine conformance to the performance requirements listed in 3.12.1, 3.13, 3.14, and 3.15 (see 3.28).

4.5.20 Environmental tests. When environmental tests are specified (see 6.1), the following tests shall be a part of this specification. The performance measurements referred to in the following tests shall be as specified in 4.5.3, 4.5.4, 4.5.5, and 4.5.7 to determine conformance to the requirements in the applicable detailed specification for resolution, video output signal level, gray scale, and sync signal waveform (see 3.29).

4.5.20.1 Humidity test. The camera to be tested shall be subjected to the following:

(a) Dry the camera for 24 hours at 130°F $\pm$ 5°F; not more than 5 percent relative humidity.
(b) Set up the camera in accordance with the requirements of 4.5.3 and 4.5.3.1 for horizontal limiting resolution.
(c) Subject the camera to two 48-hour test cycles as indicated in MIL-STD-170, except that the maximum temperature shall be 130°F.
(d) Make performance measurements to determine conformance to the requirements listed in 4.5.20. The measurements are to be made during the period specified in MIL-STD-170 while the camera is exposed to the humidity conditions shown therein.

4.5.20.2 Altitude test. Test in accordance with the recommended testing procedures for low air pressure test of IEC 68-2-13(1966). Test at a pressure equivalent to the altitude in feet as specified in the applicable detailed specification and at the low temperature specified in 4.5.20.3, for a period of not less than 2 hours. During the conditioning and the recovery period, the unit shall remain on and shall be checked for any degradation in performance of the requirements listed in 4.5.20.

4.5.20.3 Temperature test. Test in accordance with the recommended testing procedures for low temperature and high temperature of IEC 68-2-1(1966) and IEC 68-2-2(1966) Section Two - Test Ab: using the operational test procedure, at temperatures equal to those specified in the applicable detailed specification. Check for any degradation in performance of the requirements listed in 4.5.20. During the recovery period, the unit shall remain on and shall be checked for normal operation during recovery period.

4.5.20.4 Vibration test. Test in accordance with the recommended testing procedures of IEC 68-2-6 (1966). Use test Procedure A without the final resonance search. Vibrate the unit over the frequency range, at the vibration amplitude and for the period specified in the applicable detailed specification. The unit shall then be inspected to ascertain that no fixed parts have become loose or damaged and that the performance requirements are maintained.

4.5.20.5 Shock test. The test shall be conducted on a sold two-inch firebench top or equivalent surface. With the camera operating and connected to a monitor, tilt up the camera through an angle of 30 degrees. Using the front edge of the camera as an axis and permit the camera to drop back freely to the horizontal. Repeat, using the rear edge of the camera as an axis. Repeat each drop three times for a total of six drops. After each drop, the camera shall continue to perform without any attention to or replacement of parts. Adjustments may be made to external controls only. Momentary failure to operate when dropped is permissable providing the camera resumes normal operation after each drop. After the shock test, the camera shall be checked to determine conformance to the performance requirements as listed in 4.5.20.

4.6 Inspection of preparation for delivery requirements. Packaging, packing, and marking of exterior containers shall be visually inspected for the defects listed in table II to determine conformance to the requirements in Section 5. One defect shall cause rejection of a unit. Rejected units may be offered again for inspection provided defects causing rejection have been corrected.

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TABLE II. Classification of Preparation for Delivery Defects

Examine	Defects
Markings (exterior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closures of container flaps, loose strapping, inadequate stapling.
	Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Unless otherwise specified (see 6.1), commercial packaging is acceptable under this specification.

5.2 Packing. Unless otherwise specified (see 6.1), the subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to assure acceptance and safe delivery by common or other carriers, at the lowest rate at the point of delivery.

5.3 Marking. In addition to marking required by the procurement documents, the shipping container shall be marked in accordance with Fed. Std. No. 123 for civil agencies and MIL-STD-129 for military agencies as applicable.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and in the applicable detailed specification and include the following information in procurement documents.

(a) Title, number, and date of this specification and the applicable detailed specification.
(b) Quantity of item required,
(c) Specify type required (see 1.2.1 and 3.1).
(d) Preproduction model, delete when not required (see 3.4).
(e) Preproduction model, specify the place of inspection (see 3.4 and 4.2.1).
(f) Packaging and packing required if other than specified (see 5.1 and 5.2).
(g) Inspection responsibility, if other than specified (see 4.1).
(h) Specify whether environmental requirements are to apply (see 3.29).
(i) Whether environmental testing shall be conducted (see 4.2.1, 4.2.2.1.4, and 4.5.20).
(j) Specify lens to be supplied with camera, when required (see 3.2.3.1).

6.2 Definitions. For the purpose of this specification, the definitions of EIA, Industrial Electronics Bulletin No. 1, Closed Circuit Television Definitions, shall apply.

6.3 The following detailed specifications have been issued for monochrome closed circuit television cameras:

W-C-1422/1B - Camera, Television, Monochrome, Type I (Closed Circuit, General Purpose).
W-C-1422/2B - Camera, Television, Monochrome, Type II (Closed Circuit, High Resolution, Selectable Scan Rates).
W-C-1422/3C - Camera, Television, Monochrome, Type III (Closed Circuit Studio Camera With Electronic Viewfinder).

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